

$$\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 1 & 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 & 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 & 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 & 1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix} * \begin{bmatrix} -1 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 & 0 & -1 & -1 & 0 \\ 0 & 1 & 1 & 0 & 0 & -1 & -1 & 0 \\ 0 & 1 & 1 & 0 & 0 & -1 & -1 & 0 \\ 0 & 1 & 1 & 0 & 0 & -1 & -1 & 0 \\ 0 & 0 & 0 & 0 & 0 & -1 & -1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\begin{bmatrix}
 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
 0 & 1 & 1 & 0 & 0 & -1 & -1 & 0 \\
 0 & 1 & 1 & 0 & 0 & -1 & -1 & 0 \\
 0 & 1 & 1 & 0 & 0 & -1 & -1 & 0 \\
 0 & 1 & 1 & 0 & 0 & -1 & -1 & 0 \\
 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0
 \end{bmatrix}
 * \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}$$

$$\begin{bmatrix}
 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
 0 & 1 & 1 & 0 & 0 & -1 & -1 & 0 \\
 0 & 3 & 3 & 0 & 0 & -3 & -3 & 0 \\
 0 & 4 & 4 & 0 & 0 & -4 & -4 & 0 \\
 0 & 4 & 4 & 0 & 0 & -4 & -4 & 0 \\
 0 & 3 & 3 & 0 & 0 & -3 & -3 & 0 \\
 0 & 1 & 1 & 0 & 0 & -1 & -1 & 0 \\
 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0
 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 0 & 0 & 6 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 6 & 6 & 0 & 0 & 0 \\ 0 & 0 & 1 & 1 & 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 & 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 & 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 & 1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 6 & 6 & 6 & 0 & 6 & 0 \end{bmatrix}$$

$$\begin{bmatrix} -1 & -2 & -1 \\ 0 & 0 & 0 \\ 1 & 2 & 1 \end{bmatrix} = \overset{3 \times 1}{\begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}} * \overset{1 \times 3}{\begin{bmatrix} 1 & 2 & 1 \end{bmatrix}}$$

$$\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 1 & 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 & 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 & 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 & 1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix} \times \begin{bmatrix} -1 \\ -1 \\ 0 \\ -1 \end{bmatrix} =$$

$$\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 1 & 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 & 1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -1 & -1 & -1 & -1 & 0 & 0 \\ 0 & 0 & -1 & -1 & -1 & -1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 1 & 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 & 1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -1 & -1 & -1 & -1 & 0 & 0 \\ 0 & 0 & -1 & -1 & -1 & -1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix} * \begin{bmatrix} 1 & 2 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 3 & 4 & 4 & 3 & 1 & 0 \\ 0 & 1 & 3 & 4 & 4 & 3 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -1 & -3 & -4 & -4 & -3 & -1 & 0 \\ 0 & -1 & -3 & -4 & -4 & -3 & -1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\begin{bmatrix}
 0 & 1 & 1 & 0 & 0 & -1 & -1 & 0 \\
 0 & 3 & 3 & 0 & 0 & -3 & -3 & 0 \\
 0 & 4 & 4 & 0 & 0 & -4 & -4 & 0 \\
 0 & 9 & 9 & 0 & 0 & -4 & -4 & 0 \\
 0 & 3 & 3 & 0 & 0 & -3 & -3 & 0 \\
 0 & 1 & 1 & 0 & 0 & -1 & -1 & 0 \\
 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0
 \end{bmatrix}
 \begin{bmatrix}
 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
 0 & 1 & 3 & 4 & 4 & 3 & 1 & 0 \\
 0 & 1 & 3 & 4 & 4 & 3 & 1 & 0 \\
 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
 0 & -1 & -3 & -4 & -4 & -3 & -1 & 0 \\
 0 & -1 & -3 & -4 & -4 & -3 & -1 & 0 \\
 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0
 \end{bmatrix}$$

I_x
 I_y

$$F(i, j) = \sqrt{I_x^2 + I_y^2}$$

$$\begin{matrix} & F & \\ \begin{bmatrix}
 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
 0 & \sqrt{2} & \sqrt{10} & 4 & 4 & \sqrt{14} & \sqrt{2} & 0 \\
 0 & \sqrt{10} & \sqrt{18} & 4 & 4 & \sqrt{18} & \sqrt{10} & 0 \\
 0 & 4 & 4 & 0 & 0 & 4 & 4 & 0 \\
 0 & 4 & 4 & 0 & 0 & 4 & 4 & 0 \\
 0 & \sqrt{10} & \sqrt{18} & 4 & 4 & \sqrt{18} & \sqrt{10} & 0 \\
 0 & \sqrt{2} & \sqrt{10} & 4 & 4 & \sqrt{10} & \sqrt{2} & 0 \\
 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0
 \end{bmatrix}
 \end{matrix}$$